Based on Previous Years' Papers (2007-2025)

CHAPTER 1: FORCE, WORK, POWER AND ENERGY

Q1. A body is acted upon by two equal and opposite forces, that are NOT along the same straight line. The body will: [2025, 1 mark]

(a) remain stationary
(b) have only rotational motion
(c) have only rectilinear motion
(d) have both rectilinear and rotational motion

Answer: (b) have only rotational motion

Explanation: Forces are equal and opposite, but not along the same straight line. So, the forces will form a couple and the body will have rotational motion only.

Q2. Which among the following is a vector quantity? [2025, 1 mark]

(a) work
(b) power
(c) energy
(d) moment of couple

Answer: (d) moment of couple

Explanation: Vector quantity has magnitude and direction. The direction of a moment of couple is perpendicular to the plane containing the two forces forming the couple.

Q3. Clockwise moment produced by a force about a fulcrum is considered to be: [2023, 1 mark]

(a) Positive
(b) Negative
(c) Zero
(d) None of these

Answer: (b) Negative

Explanation: By convention, anticlockwise moment is considered positive and clockwise moment is considered negative.

Q4. When the speed of a moving object is doubled, then its kinetic energy: [2023, 1 mark]

(a) remains the same
(b) decreases
(c) is doubled
(d) becomes four times

Answer: (d) becomes four times

Explanation: Kinetic energy = ½ mv^2. So, kinetic energy becomes four times if the speed is doubled.

Q5. The energy conversion in a washing machine is from: [2023, 1 mark]

(a) magnetic to electrical
(b) electrical to mechanical
(c) electrical to magnetic
(d) mechanical to electrical

Answer: (b) electrical to mechanical

Explanation: In washing machine, there is a motor which converts electrical energy into mechanical energy.

Q6. The energy change during photosynthesis in plants is: [2022 Sem 1, 1 mark]

(a) heat to chemical
(b) light to chemical
(c) chemical to light
(d) chemical to heat

Answer: (b) light to chemical

Explanation: During photosynthesis, plants convert solar energy into glucose. Hence, it is light to chemical energy conversion.

Q7. A coolie raises a load upwards against the force of gravity then the work done by the load is: [2022 Sem 1, 1 mark]

(a) zero
(b) positive work
(c) negative work
(d) none of these

Answer: (c) negative work

Explanation: $W = Fs \cos \theta$. Here $\theta = 180°$, so the work done = -Fs.

Q8. Which one of the following is the correct mathematical relation? [2022 Sem 1, 1 mark]

(a) Power = Force/Velocity
(b) Power = Force × Acceleration
(c) Power = Force/Acceleration
(d) Power = Force × Velocity

Answer: (d) Power = Force × Velocity

Q9. 1 joule = _____ erg [2022 Sem 1, 1 mark]

(a) 10^9
(b) 10^7

(c) 10^5
(d) 10^6

Answer: (b) 10^7

Explanation: 1 Joule = 1 Newton × 1 metre = 10^5 dynes × 10^2 cm = 10^7 erg.

Q10. A light body A and a heavy body B have the same momentum. Choose the correct statement: [2022 Sem 1, 1 mark]

(a) kinetic energy of body A and body B will be the same
(b) kinetic energy of body A is greater than kinetic energy of body B
(c) kinetic energy of body B is greater than kinetic energy of body A
(d) unless we know the velocity, we cannot find which body has greater kinetic energy

Answer: (b) kinetic energy of body A is greater than kinetic energy of body B

Explanation: K.E. = $p^2/2m$. Since momentum is same, K.E. $\propto$ 1/m. As mB > mA, K.E.A > K.E.B.

Q11. For the above question, if the ratio of kinetic energies of A and B is 5:2, then the mass ratio of the bodies respectively is: [2022 Sem 1, 1 mark]

(a) 5:2
(b) 25:4
(c) 2:5
(d) 4:24

Answer: (c) 2:5

Explanation: K.E.A/K.E.B = mB/mA ⇒ 5/2 = mB/mA ⇒ mA/mB = 2/5.

Q12. A woman draws water from a well using a fixed pulley. The mass of the bucket and the water together is 10 kg. The force applied by the woman is 200 N. The mechanical advantage is (g = 10 m/s^2): [2022 Sem 1, 1 mark]

(a) 2
(b) 20
(c) 0.05
(d) 0.5

Answer: (d) 0.5

Explanation: Load = mg = 10 × 10 = 100 N, Effort = 200 N, M.A. = Load/Effort = 100/200 = 0.5.

Q13. A single fixed pulley is used because: [2022 Sem 1, 1 mark]

(a) it changes the direction of applied effort conveniently
(b) it multiplies speed
(c) it multiplies effort
(d) its efficiency is 100%

Answer: (a) it changes the direction of applied effort conveniently

---

Q14. In the diagram shown below, the velocity ratio of the arrangement is: [2022 Sem 1, 1 mark]

(a) 1
(b) 2
(c) 3
(d) 0

Answer: (b) 2

Explanation: If pulley B is moved by x distance downwards, then the rope moves 2x. Hence, VR = 2.

Q15. A body of mass 200 g falls freely from a height of 15 m. [g = 10 m/s²]
When the body reaches 10 m above the ground, its potential energy will be: [2022 Sem 1, 1 mark]

(a) 20000 J
(b) 10 J
(c) 10000 J
(d) 20 J

Answer: (d) 20 J

Explanation: P.E. = mgh = $200 \times 10^{-3} \times 10 \times 10 = 20$ J.

Q16. For the above question, the gain in kinetic energy of the body when it reaches 10 m above the ground is: [2022 Sem 1, 1 mark]

(a) 20 J
(b) 10 J
(c) 30 J
(d) 25 J

Answer: (b) 10 J

Explanation: At 15 m height, total energy = mgh = $200 \times 10^{-3} \times 10 \times 15 = 30$ J. At 10 m, P.E. = 20 J, so K.E. = 30 - 20 = 10 J.

Q17. For the above question, the total mechanical energy it will possess when it is just about to strike the ground is: [2022 Sem 1, 1 mark]

(a) 30000 J
(b) 20000 J
(c) 30 J
(d) 20 J

Answer: (c) 30 J

Explanation: Total energy is conserved. When the body is about to touch the ground, total energy = 30 J (all kinetic).

Q18. For the above question, the velocity in m/s with which the body will hit the ground is: [2022 Sem 1, 1 mark]

(a) 30
(b) 10
(c) $10\sqrt{3}$
(d) $10\sqrt{2}$

Answer: (c) $10\sqrt{3}$

Explanation: $\frac{1}{2} mv^2 = 30 \Rightarrow \frac{1}{2} \times 0.2 \times v^2 = 30 \Rightarrow v^2 = 300 \Rightarrow v = 10\sqrt{3}$ m/s.

Q19. A force is applied on (i) a non-rigid body and (ii) a rigid body. How does the effect of the force differ in the above two cases? [2014, 2 marks]

(a) Rigid body changes shape, non-rigid body moves
(b) Non-rigid body changes shape, rigid body moves
(c) Both change shape and move
(d) Neither changes shape nor moves

Answer: (b) Non-rigid body changes shape, rigid body moves

Explanation: A force on a rigid object does not change its dimensions but causes motion. A force on a non-rigid object changes its dimensions.

Q20. (i) What is the weight of a body placed at the centre of the earth? (ii) What is the principle of an ideal machine? [2014, 2 marks]

(a) Zero; Work output = Work input
(b) Infinite; Work output > Work input
(c) Same as on surface; Work output < Work input
(d) Zero; Work output < Work input

Answer: (a) Zero; Work output = Work input

Q21. Is it possible to have an accelerated motion with a constant speed? Explain. [2014, 2 marks]

(a) No, acceleration requires change in speed
(b) Yes, in circular motion

(c) Yes, in linear motion with changing direction
(d) No, speed and acceleration are independent

Answer: (b) Yes, in circular motion

Explanation: In uniform circular motion, velocity changes due to change in direction, hence it is accelerated even though speed is constant.

Q22. (i) When does a force do work? (ii) What is the work done by the moon when it revolves around the earth? [2014, 2 marks]

(a) When displacement occurs; Positive work
(b) When force is applied; Zero work
(c) When displacement occurs in direction of force; Zero work
(d) When force is perpendicular to displacement; Maximum work

Answer: (c) When displacement occurs in direction of force; Zero work

Explanation: Work is done when force causes displacement. The moon revolves perpendicular to gravitational force, so work done = 0.

Q23. The ratio of amplitudes of two waves is 3 : 4. The ratio of their loudness is: [2015, 2 marks]

(a) 3 : 4
(b) 4 : 3
(c) 9 : 16
(d) 16 : 9

Answer: (c) 9 : 16

Explanation: Loudness $\propto$ (amplitude)^2. So ratio = $3^2 : 4^2$ = 9 : 16.

Q24. Two bodies A and B have masses in the ratio 5:1 and their kinetic energies are in the ratio 125:9. The ratio of their velocities is: [2019, 2 marks]

(a) 5 : 3
(b) 3 : 5
(c) 25 : 9
(d) 9 : 25

Answer: (a) 5 : 3

Explanation: K.E. = ½ mv^2. $K_1/K_2 = (m_1/m_2)(v_1/v_2)^2 \Rightarrow 125/9 = (5/1)(v_1/v_2)^2 \Rightarrow (v_1/v_2)^2 = 25/9 \Rightarrow v_1/v_2 = 5/3$.

Q25. A body of mass 10 kg is kept at a height of 5 m. It is allowed to fall and reach the ground. What is the total mechanical energy possessed by the body at the height of 2 m assuming it is a frictionless medium? ($g = 10\ m/s^2$) [2019, 3 marks]

(a) 200 J
(b) 300 J
(c) 400 J
(d) 500 J

Answer: (d) 500 J

Explanation: Total mechanical energy is conserved = P.E. at maximum height = mgh = 10 × 10 × 5 = 500 J.

Q26. For the above question, what is the kinetic energy possessed by the body just before hitting the ground? [2019, 3 marks]

(a) 200 J
(b) 300 J
(c) 400 J
(d) 500 J

Answer: (d) 500 J

Explanation: Just before hitting ground, P.E. = 0, so K.E. = Total energy = 500 J.

CHAPTER 2: LIGHT - REFRACTION AND LENSES

Q1. The deviation produced by an equilateral prism does not depend on: [2022 Sem 1, 1 mark]

(a) the angle of incidence
(b) the size of the prism
(c) the material of the prism
(d) the colour of light used

Answer: (b) the size of the prism

Explanation: $\delta = i + e - A$. i depends on material and colour. A is the angle of prism. So, deviation does not depend on the size of prism.

Q2. The refractive index of a diamond is 2.4. It means that: [2022 Sem 1, 1 mark]

(a) the speed of light in vacuum is equal to 2.4 times the speed of light in diamond
(b) the speed of light in the diamond is 2.4 times the speed of light in a vacuum
(c) the speed of light in a vacuum is 2.4 times the speed of light in the diamond
(d) the wavelength of light in diamond is 2.4 times the wavelength of light in vacuum

Answer: (c) the speed of light in a vacuum is 2.4 times the speed of light in the diamond

Explanation: $\mu = c/v \Rightarrow 2.4 = c/v \Rightarrow c = 2.4v$.

Q3. An object of height 10 cm is placed in front of a concave lens of focal length 20 cm at a distance 25 cm from the lens. Is it possible to capture this image on a screen? [2022 Sem 1, 1 mark]

(a) Yes, as the image formed will be real
(b) Yes, as the image formed will be erect
(c) No, as the image formed will be virtual
(d) No, as the image formed will be inverted

Answer: (c) No, as the image formed will be virtual

Explanation: Concave lens always produces virtual image.

Q4. The colour of white light which is deviated least by a prism is: [2022 Sem 1, 1 mark]

(a) green
(b) yellow
(c) red
(d) violet

Answer: (c) red

Explanation: Red deviates least since its wavelength is the largest.

Q5. The wavelength range of visible light is: [2022 Sem 1, 1 mark]

(a) 40 nm to 80 nm
(b) 4000 nm to 8000 nm
(c) 4 nm to 8 nm
(d) 400 nm to 800 nm

Answer: (d) 400 nm to 800 nm

Q6. Observe the diagram which shows the path of an incident ray through an optical plane LL' of a lens. The focal length of the lens is 20 cm. If an object is placed at a distance of 30 cm in front of this lens, then: [2022 Sem 1, 1 mark]

(a) the image will be virtual
(b) the image will be diminished and inverted
(c) the image will be diminished
(d) the image will be real and magnified

Answer: (d) the image will be real and magnified

Explanation: The lens is convex. If object is placed between F and 2F, real, inverted and magnified image is formed.

Q7. For the above lens, this type of lens can be used: [2022 Sem 1, 1 mark]

(a) to correct hypermetropia
(b) to correct myopia
(c) to diverge light
(d) in the front door peepholes

Answer: (a) to correct hypermetropia

Explanation: In hypermetropia, convex lens is used so that rays meet on the retina.

Q8. An object is placed in front of this lens at a distance of 60 cm. Then the image distance from the lens with proper sign convention is: [2022 Sem 1, 1 mark]

(a) +60 cm
(b) +30 cm
(c) -30 cm
(d) +15 cm

Answer: (b) +30 cm

Explanation: Using lens formula, $1/v - 1/u = 1/f \Rightarrow 1/v + 1/60 = 1/20 \Rightarrow 1/v = 1/30 \Rightarrow v = +30$ cm.

Q9. For the above, the magnification of the image is: [2022 Sem 1, 1 mark]

(a) 0.25
(b) 1.25
(c) -0.5
(d) 1

Answer: (c) -0.5

Explanation: Magnification = $v/u = 30/(-60) = -0.5$.

Q10. The diagram below shows a ray of light travelling from air into a glass material. The angle of incidence at the surface AB is: [2022 Sem 1, 1 mark]

(a) 43°
(b) 47°
(c) 90°
(d) 0°

Answer: (b) 47°

Explanation: Angle of incidence is the angle between incident ray and normal = 90° - 43° = 47°.

Q11. For the above diagram, select the correct statement: [2022 Sem 1, 1 mark]

(a) The speed of light at the curved surface AD does not change while entering the block
(b) The ray at the surface AD is not travelling along the radius of the curved part
(c) The ray at the surface AD is travelling along the radius of the curved part
(d) Light never refracts when it enters a curved surface

Answer: (c) The ray at the surface AD is travelling along the radius of the curved part

---

Q12. For the above diagram, the angle of incidence on the surface BC is: [2022 Sem 1, 1 mark]

(a) 43°
(b) 47°
(c) 90°
(d) 0°

Answer: (a) 43°

Q13. For the above diagram, the critical angle of this material of glass is: [2022 Sem 1, 1 mark]

(a) 47°
(b) 43°
(c) 42°
(d) 45°

Answer: (b) 43°

Q14. The diagram below shows the path of light passing through a right-angled prism of critical angle 42°. The angle C of the prism is: [2022 Sem 1, 1 mark]

(a) 45°
(b) 60°
(c) 90°
(d) 30°

Answer: (d) 30°

Q15. For the above prism, which one of the following diagrams shows the correct path of this ray till it emerges out of the prism? [2022 Sem 1, 1 mark]

(a) Option A
(b) Option B
(c) Option C
(d) Option D

Answer: (b) Option B

Q16. Speed of blue light in water is: [2023, 1 mark]

(a) more than green light
(b) more than orange light
(c) more than violet light
(d) more than red light

Answer: (c) more than violet light

Explanation: Refractive index for blue light is less than that for violet light. Since $v = c/\mu$, velocity of blue light is more than that of violet light.

Q17. A concave lens produces only image: [2023, 1 mark]

(a) real, enlarged
(b) virtual, enlarged
(c) virtual, diminished
(d) real, diminished

Answer: (c) virtual, diminished

Q18. When a ray of light travels normal to the given surface, then the angle of refraction is: [2023, 1 mark]

(a) 180°
(b) 90°
(c) 0°
(d) 45°

Answer: (c) 0°

Explanation: The ray travels undeviated. Hence angle of incidence = angle of refraction = 0°.

Q19. Small air bubbles rising up a fish tank appear silvery when viewed from some particular angle is due to: [2023, 1 mark]

(a) reflection
(b) refraction
(c) dispersion
(d) total internal reflection

Answer: (d) total internal reflection

Q20. The ratio of velocities of light of wavelength 400 nm and 800 nm in a vacuum is: [2022 Sem 1, 1 mark]

(a) 1:1
(b) 1:2
(c) 2:1
(d) 1:3

Answer: (a) 1:1

Explanation: Velocity of light in vacuum is same for all wavelengths.

CHAPTER 3: SOUND

Q1. When the stem of vibrating tuning fork is pressed on a table, the tabletop starts vibrating. These vibrations are definitely an example of: [2024, 1 mark]

(a) resonance
(b) natural vibrations
(c) forced vibrations
(d) damped vibrations

Answer: (c) forced vibrations

Q2. The human ear can detect continuous sounds in the frequency range from 20 Hz to 20,000 Hz. Assuming speed of sound in air is 330 m/s for all frequencies, the wavelengths corresponding to the extreme frequencies are: [2017, 2 marks]

(a) 16.5 m and 16.5 mm
(b) 16.5 m and 1.65 cm
(c) 16.5 cm and 16.5 mm
(d) 165 m and 1.65 cm

Answer: (a) 16.5 m and 16.5 mm

Explanation: $\lambda_{max} = v/f_{min} = 330/20 = 16.5$ m; $\lambda_{min} = v/f_{max} = 330/20000 = 0.0165$ m = 16.5 mm.

Q3. An enemy plane is at a distance of 300 km from a radar. The time taken by radar to detect the plane (velocity of radio waves = 3×10^8 m/s) is: [2017, 2 marks]

(a) 2×10^{-3} s
(b) 2×10^{-2} s
(c) 2×10^{-4} s
(d) 2 s

Answer: (a) 2×10^{-3} s

Explanation: $t = 2d/v = 2 \times 300 \times 10^3 / (3 \times 10^8) = 2 \times 10^{-3}$ s.

Q4. How is the frequency of a stretched string related to its length and tension? [2017, 2 marks]

(a) $f \propto l$, $f \propto \sqrt{T}$
(b) $f \propto 1/l$, $f \propto \sqrt{T}$
(c) $f \propto l$, $f \propto T$
(d) $f \propto 1/l$, $f \propto T$

Answer: (b) $f \propto 1/l$, $f \propto \sqrt{T}$

Q5. Free vibrations are: [2022 Sem 2, 1 mark]

(a) the vibrations under the influence of a periodic force
(b) the vibrations with larger amplitude
(c) the vibrations when the frequency continuously decreases
(d) the vibrations with a constant frequency and constant amplitude

Answer: (d) the vibrations with a constant frequency and constant amplitude

---

Q6. The diagram below shows four sound waves. Which sound has the highest pitch? [2022 Sem 2, 1 mark]

(a) Wave A
(b) Wave B
(c) Wave C
(d) Wave D

Answer: (b) Wave B

Explanation: Pitch depends on frequency. Wave B has the highest frequency.

Q7. Select a correct option with respect to echo depth sounding: [2022 Sem 1, 1 mark]

(a) infrasonic waves are used
(b) the frequency of the waves used is between 20 Hz and 20,000 Hz
(c) ultrasonic waves are used
(d) supersonic waves are used

Answer: (c) ultrasonic waves are used

Q8. Which one of the following diagnostic methods use reflection of sound? [2022 Sem 1, 1 mark]

(a) CT scan
(b) Electrocardiogram
(c) Echo cardiogram
(d) MRI

Answer: (c) Echo cardiogram

Q9. A boy standing in front of a wall produces two whistles per second. He notices that the sound of his whistling coincides with the echo. The time in which the boy hears the echo is: [2022 Sem 1, 1 mark]

(a) 1 s
(b) 0.5 s
(c) 1.5 s
(d) 2 s

Answer: (b) 0.5 s

Explanation: Since sound of whistling coincides with echo, the reflected sound returns in 0.5 s.

Q10. For the above, the distance at which the boy is standing from the wall (speed of sound = 320 m/s) is: [2022 Sem 1, 1 mark]

(a) 160 m
(b) 240 m
(c) 320 m
(d) 80 m

Answer: (d) 80 m

Explanation: $v = 2d/t \Rightarrow 320 = 2d/0.5 \Rightarrow d = 80$ m.

Q11. If the speed of sound is increased by 16 m/s and the boy moves 4 m away from the wall, then the time to hear echo is: [2022 Sem 1, 1 mark]

(a) 0.525 s
(b) 0.5 s
(c) 0.48 s
(d) 0.3 s

Answer: (b) 0.5 s

Explanation: New $v = 336$ m/s, new $d = 84$ m, $t = 2d/v = 168/336 = 0.5$ s.

Q12. In which of the following timings of reflection, the echo cannot be heard? [2022 Sem 1, 1 mark]

(a) 0.05 s
(b) 0.12 s
(c) 0.2 s
(d) 0.11 s

Answer: (a) 0.05 s

Explanation: To hear an echo, time difference should be ≥ 0.1 s.

Q13. A man standing in front of a vertical cliff fires a gun. He hears the echo after 3.5 seconds. On moving closer to the cliff by 84 m, he hears the echo after 3 seconds. The distance of the cliff from the initial position is: [2020, 4 marks]

(a) 588 m
(b) 500 m
(c) 600 m
(d) 650 m

Answer: (a) 588 m

Explanation: $v = 2d/t$. Since velocity is same, $2d/3.5 = 2(d-84)/3 \Rightarrow d = 588$ m.

CHAPTER 4: ELECTRICITY AND MAGNETISM

Q1. An electric bulb of resistance 500 Ω draws a current of 0.4 A. The power of the bulb and the potential difference at its ends are: [2017, 2 marks]

(a) 80 W, 200 V
(b) 200 W, 80 V
(c) 40 W, 100 V
(d) 100 W, 40 V

Answer: (a) 80 W, 200 V

Explanation: $P = I^2R = 0.4^2 \times 500 = 80$ W; $V = IR = 0.4 \times 500 = 200$ V.

Q2. State two causes of energy loss in a transformer: [2017, 2 marks]

(a) Copper loss and hysteresis loss
(b) Iron loss and eddy current loss
(c) Both (a) and (b)
(d) Only (a)

Answer: (c) Both (a) and (b)

Q3. The graph plotted for potential difference (V) against current (I) for ohmic resistors is: [2022 Sem 2, 1 mark]

(a) A curve passing through the origin
(b) A straight line not passing through origin
(c) A straight line passing through the origin
(d) A circle centred at the origin

Answer: (c) A straight line passing through the origin

Explanation: $I \propto V$, so it is a straight line through origin.

Q4. A main switch in the main distribution board is present in: [2022 Sem 2, 1 mark]

(a) a live wire
(b) a neutral wire
(c) a live as well as neutral wire
(d) an earth wire

Answer: (c) a live as well as neutral wire

Q5. A conductor AB is kept along north-south direction of the earth above a magnetic needle. When the key K is closed, then: [2022 Sem 2, 1 mark]

(a) the needle will not show any deflection
(b) the needle will deflect towards east
(c) the needle will turn in the opposite direction i.e., towards south
(d) the needle will deflect towards west

Answer: (b) the needle will deflect towards east

Q6. A coil wound around a piece of soft iron can become an electromagnet only when: [2022 Sem 2, 1 mark]

(a) the circuit is open
(b) a magnetic compass is present in the vicinity
(c) a galvanometer is connected to the circuit
(d) a current flows in the circuit

Answer: (d) a current flows in the circuit

Q7. When a current I flows through a wire of resistance R for time t, the electrical energy produced is given by: [2023, 1 mark]

(a) I^2Rt
(b) IR^2t
(c) IRt
(d) IR^2

Answer: (a) I^2Rt

Q8. Choose the correct relation for e.m.f. (ε) and terminal voltage V: [2023, 1 mark]

(a) $\varepsilon = V$ (always)
(b) $V > \varepsilon$ (always)
(c) $V < \varepsilon$ (When the cell is in use)
(d) None of these

Answer: (c) $V < \varepsilon$ (When the cell is in use)

Explanation: $\varepsilon = V + Ir$. When $i \neq 0$, $\varepsilon > V$.

Q9. If the strength of the current flowing through a wire is increased, the strength of the magnetic field produced by it: [2023, 1 mark]

(a) decreases
(b) increases
(c) remains the same
(d) first increases then decreases

Answer: (b) increases

Q10. A geyser is rated 240 W - 220 V. This means: [2023, 2 marks]

(a) When connected to 220 V, it consumes 240 J/s
(b) When connected to 240 V, it consumes 220 J/s
(c) It has resistance 240 Ω
(d) It draws current of 220 A

Answer: (a) When connected to 220 V, it consumes 240 J/s

Q11. An appliance rated 440 W, 220 V is connected across 220 V supply. The maximum current that the appliance can draw is: [2022 Sem 2, 3 marks]

(a) 1 A
(b) 2 A
(c) 4 A
(d) 0.5 A

Answer: (b) 2 A

Explanation: $I = P/V = 440/220 = 2$ A.

Q12. For the above appliance, the resistance of the appliance is: [2022 Sem 2, 3 marks]

(a) 110 Ω
(b) 220 Ω
(c) 55 Ω
(d) 440 Ω

Answer: (a) 110 Ω

Explanation: $R = V^2/P = 220^2/440 = 110$ Ω.

Q13. If a wire of resistance 2 Ω gets stretched to thrice its original length, the new resistance is: [2022 Sem 2, 4 marks]

(a) 6 Ω
(b) 12 Ω
(c) 18 Ω
(d) 9 Ω

Answer: (c) 18 Ω

Explanation: $R' = n^2R = 9 \times 2 = 18$ Ω.

Q14. For the above wire, what happens to the specific resistance? [2022 Sem 2, 4 marks]

(a) Increases 9 times
(b) Decreases 9 times
(c) Remains same
(d) Increases 3 times

Answer: (c) Remains same

Explanation: Specific resistance depends on material, not dimensions.

Q15. In a step-up transformer, the turns ratio is: [2025, 1 mark]

(a) more than 1
(b) less than 1
(c) equal to 1
(d) zero

Answer: (a) more than 1

CHAPTER 5: HEAT AND CALORIMETRY

Q1. Define heat capacity and state its S.I. unit: [2017, 2 marks]

(a) Energy to raise temperature by 1°C; J/K
(b) Energy to raise temperature by 1°C; J/°C
(c) Energy per unit mass; J/kg°C
(d) Energy per mole; J/mol

Answer: (a) Energy to raise temperature by 1°C; J/K

Q2. Why is the base of a cooking pan generally made thick? [2017, 2 marks]

(a) To increase heat capacity for slow heating
(b) To decrease heat capacity for quick heating
(c) To increase conductivity
(d) To decrease weight

Answer: (a) To increase heat capacity for slow heating

Q3. A solid of mass 50 g at 150°C is placed in 100 g of water at 11°C. The final temperature is 20°C. The specific heat capacity of the solid is (sp. heat of water = 4.2 J/g°C): [2017, 2 marks]

(a) 0.582 J/g°C
(b) 0.4 J/g°C
(c) 0.8 J/g°C
(d) 1.2 J/g°C

Answer: (a) 0.582 J/g°C

Explanation: Heat lost = Heat gained ⇒ 50 × c × (150-20) = 100 × 4.2 × (20-11) ⇒ 6500c = 3780 ⇒ c = 0.582 J/g°C.

Q4. If water absorbs 4000 joule heat to increase the temperature of 1 kg water through 1°C, then the specific heat capacity of water is: [2022 Sem 2, 1 mark]

(a) 4 J/kg°C
(b) 400 J/g°C
(c) 4 J/g°C
(d) 4.2 J/g°C

Answer: (c) 4 J/g°C

Explanation: 4000 J for 1 kg (1000 g) through 1°C ⇒ sp. heat = 4 J/g°C.

Q5. Water is used in car radiators because: [2022 Sem 2, 1 mark]

(a) it is a good conductor of heat
(b) it conducts heat faster than other substances
(c) its specific heat capacity is very low
(d) its specific heat capacity is very high so it can cool engine without greater increase in its own temperature

Answer: (d) its specific heat capacity is very high so it can cool engine without greater increase in its own temperature

---

Q6. Specific latent heat of a substance: [2023, 1 mark]

(a) is directly proportional to the mass
(b) is directly proportional to the change in temperature
(c) depends on the material
(d) is inversely proportional to the mass

Answer: (c) depends on the material

Q7. Specific heat capacity of a substance X is 1900 $Jkg^{-1}°C^{-1}$ means: [2023, 1 mark]

(a) Substance X absorbs 1900 J for 1°C rise in temperature
(b) 1 kg of substance X absorbs 1900 J heat for 1°C rise in temperature
(c) 1 kg of substance X absorbs 1900 J heat to increase the temperature
(d) 1 kg of substance X absorbs 1900 J heat to cool down by 1°C

Answer: (b) 1 kg of substance X absorbs 1900 J heat for 1°C rise in temperature

---

Q8. Calculate the quantity of heat produced in a 20 Ω resistor carrying 2.5 A current in 5 minutes: [2016, 2 marks]

(a) 37500 J
(b) 3750 J
(c) 375 J
(d) 375000 J

Answer: (a) 37500 J

Explanation: $H = I^2Rt = (2.5)^2 \times 20 \times (5 \times 60) = 6.25 \times 20 \times 300 = 37500$ J.

Q9. Calculate the mass of ice required to lower the temperature of 300 g of water at 40°C to water at 0°C (sp. latent heat of ice = 336 J/g, sp. heat of water = 4.2 J/g°C): [2016, 2 marks]

(a) 100 g
(b) 150 g
(c) 200 g
(d) 250 g

Answer: (b) 150 g

Explanation: Heat lost by water = $300 \times 4.2 \times 40 = 50400$ J. Heat gained by ice = $m \times 336$. So, $m = 50400/336 = 150$ g.

Q10. 85 g of water at 30°C is cooled to 5°C by adding certain mass of ice. Find the mass of ice required (sp. heat of water = 4.2 J/g°C, sp. latent heat of fusion = 336 J/g): [2024, 3 marks]

(a) 20 g
(b) 25 g
(c) 30 g
(d) 35 g

Answer: (b) 25 g

Explanation: $85 \times 4.2 \times (30-5) = m \times 336 + m \times 4.2 \times 5 \Rightarrow 8925 = 357m \Rightarrow m = 25$ g.

CHAPTER 6: MODERN PHYSICS - RADIOACTIVITY

Q1. A radioactive nucleus containing 128 nucleons emits a β-particle. After β-emission, the number of nucleons present in the nucleus will be: [2024, 1 mark]

(a) 128
(b) 129
(c) 124
(d) 127

Answer: (a) 128

Explanation: Number of nucleons (mass number) does not change due to β emission. Only atomic number increases by 1.

Q2. Which of the following radiations suffer maximum deflection in a magnetic field? [2023, 1 mark]

(a) Alpha radiations
(b) Beta radiations
(c) Gamma radiations
(d) X-radiations

Answer: (b) Beta radiations

Explanation: β particles are fast moving negatively charged particles (electrons) with very high e/m ratio.

Q3. The heaviest nuclear radiation is: [2022 Sem 2, 1 mark]

(a) X-radiation
(b) α-radiation
(c) γ-radiation
(d) β-radiation

Answer: (b) α-radiation

Explanation: α radiations are basically helium nucleus, hence heaviest.

Q4. To study the age of excavated material of archaeological significance, we study the rate of decay of an isotope of: [2022 Sem 2, 1 mark]

(a) Uranium
(b) Cobalt
(c) Carbon
(d) Chlorine

Answer: (c) Carbon

Explanation: Radiocarbon dating uses radioactive isotope of carbon (C-14).

Q5. A radioactive element P emits one α-particle and transforms to a new element Q. What will be the position of the element Q in the periodic table? [2025, 1 mark]

(a) One group to the left of P
(b) One group to the right of P
(c) Two groups to the right of P
(d) Two groups to the left of P

Answer: (d) Two groups to the left of P

Explanation: Atomic number of Q is 2 less than that of P. Hence position will be 2 groups left of P.

Q6. The nuclear radiation with lowest ionising power is: [2025, 1 mark]

(a) α
(b) β
(c) γ
(d) X-rays

Answer: (c) γ

Q7. After emission of a nuclear radiation, the atomic number of the daughter nucleus increases by 1. Identify the nuclear radiation: [2022 Sem 2, 4 marks]

(a) Alpha
(b) Beta
(c) Gamma
(d) Neutron

Answer: (b) Beta

Q8. A radioactive substance is oxidized. Will there be any change in the nature of its radioactivity? [2016, 2 marks]

(a) Yes, it increases
(b) Yes, it decreases
(c) No, because radioactivity is a nuclear phenomenon
(d) No, because oxidation is reversible

Answer: (c) No, because radioactivity is a nuclear phenomenon

---

Q9. A nucleus

84202X

84

202

X of an element emits an alpha particle followed by a beta particle. The final nucleus is

aYb

a

Y

b

. The values of a and b are: [2020, 2 marks]

(a) 83, 198
(b) 82, 198
(c) 83, 202
(d) 84, 202

Answer: (a) 83, 198

Explanation: After α decay: mass number decreases by 4, atomic number decreases by 2. After β decay: atomic number increases by 1, mass number unchanged. So final: a = 83, b = 198.

Q10. An atomic nucleus A is composed of 84 protons and 128 neutrons. The nucleus A emits an alpha particle and is transformed into nucleus B. The composition of B is: [2020, 4 marks]

(a) 82 protons, 126 neutrons
(b) 82 protons, 124 neutrons
(c) 84 protons, 126 neutrons
(d) 86 protons, 126 neutrons

Answer: (a) 82 protons, 126 neutrons

Explanation: α particle has 2 protons and 2 neutrons. So, protons = 84-2 = 82, neutrons = 128-2 = 126.

CHAPTER 7: MACHINES AND LEVERS

Q1. Which class of lever has a mechanical advantage always greater than one? [2007, 2 marks]

(a) Class I
(b) Class II
(c) Class III
(d) All classes

Answer: (b) Class II

Q2. Which of the following is a class III lever? [2024, 1 mark]

(a) Pair of scissors
(b) Wheelbarrow
(c) Crowbar
(d) Human forearm

Answer: (d) Human forearm

Q3. For a lever, a graph is plotted with load on Y-axis and effort on X-axis. The slope of the graph represents: [2025, 1 mark]

(a) Mechanical advantage
(b) Velocity ratio
(c) Efficiency
(d) Load arm

Answer: (a) Mechanical advantage

Q4. Name a machine which can be used to: (i) multiply force (ii) change the direction of force applied: [2011, 2 marks]

(a) Inclined plane; Pulley
(b) Lever; Pulley
(c) Pulley; Lever
(d) Screw; Wedge

Answer: (a) Inclined plane; Pulley

Q5. The diagram below shows a lever in use. To which class of lever does it belong? If FA = 40 cm, AB = 60 cm, then the mechanical advantage is: [2011, 2 marks]

(a) Class I, M.A. = 2.5
(b) Class II, M.A. = 2.5
(c) Class II, M.A. = 1.5
(d) Class III, M.A. = 2.5

Answer: (b) Class II, M.A. = 2.5

Explanation: Effort arm = 40+60 = 100 cm, Load arm = 40 cm, M.A. = 100/40 = 2.5.

Q6. A uniform metre scale can be balanced at the 70.0 cm mark when a mass of 0.05 kg is hung from the 94.0 cm mark. The mass of the metre scale is: [2011, 4 marks]

(a) 0.02 kg
(b) 0.04 kg
(c) 0.06 kg
(d) 0.08 kg

Answer: (c) 0.06 kg

Explanation: Let mass of scale be M. Taking moments about 70 cm: 0.05 × (94-70) = M × (70-50) ⇒ 0.05 × 24 = M × 20 ⇒ M = 1.2/20 = 0.06 kg.

Q7. A half metre rod is pivoted at the centre with two weights of 20 gf and 12 gf suspended at perpendicular distances of 6 cm and 10 cm from the pivot respectively. Which of the two forces causes clockwise moment? [2018, 3 marks]

(a) 20 gf
(b) 12 gf
(c) Both
(d) Neither

Answer: (b) 12 gf

Q8. For the above, is the rod in equilibrium? [2018, 3 marks]

(a) Yes, because moments are equal
(b) No, because moments are unequal

(c) Yes, because forces are equal
(d) No, because forces are unequal

Answer: (a) Yes, because moments are equal

Explanation: Moment of 20 gf = 20 × 6 = 120 gf cm, Moment of 12 gf = 12 × 10 = 120 gf cm.

Q9. If the direction of 20 kgf force is reversed, the magnitude of the resultant moment on the rod is: [2018, 3 marks]

(a) 120 gf cm
(b) 240 gf cm
(c) 0 gf cm
(d) 360 gf cm

Answer: (b) 240 gf cm

Explanation: Resultant moment = 120 + 120 = 240 gf cm (both clockwise).

Q10. A block and tackle system has V.R. = 5. Rohan exerts a pull of 150 kgf. The maximum load he can raise with this pulley system if its efficiency = 75% is: [2015, 4 marks]

(a) 562.5 kgf
(b) 750 kgf
(c) 450 kgf
(d) 600 kgf

Answer: (a) 562.5 kgf

Explanation: Efficiency = M.A./V.R. ⇒ 0.75 = M.A./5 ⇒ M.A. = 3.75. Load = M.A. × Effort = 3.75 × 150 = 562.5 kgf.

CHAPTER 8: ELECTROMAGNETISM

Q1. A copper conductor is placed over two stretched copper wires whose ends are connected to a D.C. supply. To experience force in the upward direction, the magnetic poles at A and B should be: [2023, 2 marks]

(a) South at A, North at B
(b) North at A, South at B
(c) North at both
(d) South at both

Answer: (b) North at A, South at B

Q2. Name the law used to find these polarities: [2023, 2 marks]

(a) Fleming's right hand rule
(b) Fleming's left hand rule
(c) Right hand thumb rule
(d) Lenz's law

Answer: (b) Fleming's left hand rule

Q3. A copper rod PQ carrying current is kept in a magnetic field. The copper rod PQ will move towards C. State True or False: [2025, 4 marks]

(a) True
(b) False
(c) Cannot be determined
(d) Depends on current direction

Answer: (a) True

Q4. Name the law used to determine the direction of motion of PQ: [2025, 4 marks]

(a) Fleming's right hand rule
(b) Fleming's left hand rule

(c) Right hand thumb rule
(d) Lenz's law

Answer: (b) Fleming's left hand rule

Q5. What will be the effect on the force experienced, if the rod PQ is replaced by another copper rod of same length but of greater cross-sectional area? [2025, 4 marks]

(a) Force decreases
(b) Force increases
(c) Force remains same
(d) Force becomes zero

Answer: (b) Force increases

Explanation: Resistance decreases, current increases, so F = BIL increases.

Q6. State two ways by which the magnetic field of a solenoid can be made stronger: [2010, 4 marks]

(a) Increase current, increase number of turns
(b) Decrease current, decrease number of turns
(c) Increase length, decrease area
(d) Use AC instead of DC

Answer: (a) Increase current, increase number of turns

Q7. What material is used for making the armature of an electric bell? [2010, 4 marks]

(a) Steel
(b) Soft iron
(c) Copper
(d) Aluminium

Answer: (b) Soft iron

Q8. A straight wire conductor passes vertically through a piece of cardboard sprinkled with iron filings. When current is passed upward, the magnetic field lines are: [2010, 3 marks]

(a) Circular clockwise
(b) Circular anticlockwise
(c) Radial outward
(d) Radial inward

Answer: (b) Circular anticlockwise

Q9. Name the law which helps to find the direction of the magnetic field lines: [2010, 3 marks]

(a) Fleming's left hand rule
(b) Fleming's right hand rule
(c) Right hand thumb rule
(d) Lenz's law

Answer: (c) Right hand thumb rule

Q10. State two advantages of electromagnets over permanent magnets: [2018, 2 marks]

(a) Can be switched on/off, strength can be controlled
(b) Cheaper, lighter
(c) Smaller size, higher strength
(d) No energy consumption, permanent

Answer: (a) Can be switched on/off, strength can be controlled

---

CHAPTER 9: HOUSEHOLD CIRCUITS

Q1. Identify the correct option that displays the correct wiring with correct colour code: [2024, 1 mark]

(a) Blue wire (neutral) directly to appliance, brown wire (live) through switch
(b) Brown wire (neutral) directly to appliance, blue wire (live) through switch
(c) Both wires through switch
(d) Both wires directly to appliance

Answer: (a) Blue wire (neutral) directly to appliance, brown wire (live) through switch

---

Q2. A fuse is rated 5 A. Can it be used with a geyser rated 1540 W, 220 V? [2024, 2 marks]

(a) Yes, current drawn is 5 A
(b) No, current drawn is 7 A
(c) Yes, current drawn is 6 A
(d) No, current drawn is 8 A

Answer: (b) No, current drawn is 7 A

Explanation: I = P/V = 1540/220 = 7 A, which is more than 5 A rating.

Q3. According to the NEW colour convention, which colour of wire is connected to: (a) the metal body of the appliance (b) the switch of the appliance? [2025, 2 marks]

(a) Earth (green/yellow); Live (brown)
(b) Live (brown); Earth (green/yellow)
(c) Neutral (light blue); Live (brown)
(d) Earth (green/yellow); Neutral (light blue)

Answer: (a) Earth (green/yellow); Live (brown)

Q4. Three wires with proper colour coding are connected to the three terminals of a three-pin socket. Match the colour with proper terminals: [2022 Sem 2, 3 marks]

(a) Brown - Live, Green - Earth, Light blue - Neutral
(b) Brown - Earth, Green - Live, Light blue - Neutral

(c) Brown - Neutral, Green - Earth, Light blue - Live
(d) Brown - Live, Green - Neutral, Light blue - Earth

Answer: (a) Brown - Live, Green - Earth, Light blue - Neutral

---

Q5. Why is it not advisable to use a piece of copper wire as fuse wire in an electric circuit? [2020, 2 marks]

(a) Copper has low resistance and high melting point
(b) Copper has high resistance and low melting point
(c) Copper is expensive
(d) Copper corrodes easily

Answer: (a) Copper has low resistance and high melting point

Q6. Which electrical component protects the electric circuit in case of excess current and can also be used as a switch? [2023, 2 marks]

(a) Fuse
(b) MCB
(c) Rheostat
(d) Ammeter

Answer: (b) MCB

Q7. Name the wire to which this electrical component is connected in an electric circuit: [2023, 2 marks]

(a) Neutral wire
(b) Earth wire
(c) Live wire
(d) Any wire

Answer: (c) Live wire

Q8. In a household circuit, the switch should be connected to the: [2007, 2 marks]

(a) Neutral wire
(b) Earth wire
(c) Live wire
(d) Any wire

Answer: (c) Live wire

Q9. The colour code for the insulation on the earth wire is: [2011, 2 marks]

(a) Red
(b) Black
(c) Green
(d) Blue

Answer: (c) Green

Q10. Two safety devices connected to the live wire of a household electric circuit are: [2018, 3 marks]

(a) Fuse and MCB
(b) Fuse and ammeter
(c) Switch and voltmeter
(d) Earth wire and fuse

Answer: (a) Fuse and MCB